David Cropper

AI Engineer | Applied AI · Workflow Automation · Agentic AI

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PROFESSIONAL PROFILE

AI Engineer building practical AI applications and automated business workflows using Python, FastAPI, n8n and LLM APIs. Independently built workflow automations integrating email, Slack, PostgreSQL and APIs, alongside a LangGraph-based multi-agent support prototype. Additional projects cover RAG, operational intelligence and financial validation. Brings commercial enterprise Data Engineering and software testing experience, emphasising reliable integrations, deterministic controls, evaluation and human oversight.

SELECTED ENGINEERING STRENGTHS

- **Agentic workflow development:** Built a LangGraph support workflow with specialist agents, supervisor-led investigation, deterministic validation and human approval.
- **Workflow integration:** n8n orchestration across LLM APIs, Slack, Gmail, PostgreSQL, forms, webhooks and HTTP service endpoints.
- **Engineering controls:** structured outputs, SQL reconciliation, idempotency, duplicate detection, policy enforcement, failure handling and owner approval.
- **Cloud and delivery:** Python/FastAPI services, Docker, PyTest, Git and hands-on AWS migration using ECS/Fargate, ECR, S3, IAM and CloudWatch.

TECHNICAL SKILLS

AI / Agents: OpenAI APIs, LangGraph, LangChain, LLM integration, tool invocation, RAG, embeddings, ChromaDB, structured outputs, evaluation, human-in-the-loop governance

Automation / Integration: n8n, Slack, Gmail, PostgreSQL, REST APIs, webhooks, forms, conditional routing, retries, Microsoft Copilot Studio and Power Automate (hands-on prototype)

Software / Cloud: Python, FastAPI, Pydantic, SQL, Pandas, Docker, Git, PyTest, AWS (S3, ECR, ECS/Fargate, IAM, CloudWatch), Jenkins, Apache NiFi, Apache Airflow

INDEPENDENT AI ENGINEERING EXPERIENCE

AI Engineer — Independent R&D | Chase Risk & Compliance

Dec 2025 – Present

Portfolio-led independent engineering · Melbourne

Designs and develops AI-enabled applications and operational workflows, separating probabilistic interpretation from deterministic execution, validation and approval. Produces testable solutions and public project demonstrations; work is independent engineering rather than a claim of long-term commercial AI employment.

FEATURED PROJECT — GOVERNED AGENTIC SUPPORT ORCHESTRATION PLATFORM

Python · LangGraph · FastAPI · n8n · Slack · LLM APIs · mock enterprise systems

- Built an enterprise-style support workflow for a failed booking and duplicate-payment scenario, combining specialist evidence, booking/payment, policy and reconciliation agents with a supervisor that can dynamically request further investigation.

- Orchestrated bounded agent reasoning with LangGraph, exposing case initiation and workflow continuation through FastAPI while n8n coordinates Slack notifications, approval collection and final operational updates.
- Separated LLM-proposed resolutions from deterministic controls validating payment target, amount, policy fit and evidence sufficiency before a human approve/reject decision using interrupt/resume.
- Applied least-privilege evidence exposure, fail-closed behaviour, loop bounds and explicit approval authority; **refund execution is intentionally simulated/not enabled**.

APPLIED AI WORKFLOW AUTOMATION — NORTHSTAR PROJECTS

n8n · OpenAI · PostgreSQL · SQL · Slack · Gmail · HTTP APIs · simulated business systems

Designed four complementary business-process prototypes to demonstrate where LLM interpretation adds value, where exact business rules must remain deterministic, and how integrations, exception handling and approval controls fit together in operational workflows.

Northstar — Support Intake & Triage

Processes incoming customer-support email to reduce manual reading, urgency assessment and routing, while retaining manual review for ambiguous or invalid inputs.

- Integrated Gmail intake with OpenAI-based structured classification, summarisation and urgency assessment in n8n.
- Applied deterministic Switch/IF routing to appropriate Slack support channels; handled incomplete or ambiguous cases through manual-review paths.
- Added API retry behaviour, failure notifications and an error workflow; exercised positive, negative and adversarial input scenarios.

Northstar — Invoice Intake & Validation

Automates the intake of PDF invoices while keeping financial checks, deduplication and external submission decisions under conventional software control.

- Used OpenAI to extract structured invoice fields from PDFs, followed by deterministic required-field and arithmetic validation.
- Queried PostgreSQL to detect previously processed invoices; persisted idempotency keys and processing states to prevent repeat submission.
- Submitted validated payloads to a **simulated/test accounting API** and recorded failure cases with Slack exception notifications; no live accounting-system integration claimed.

Northstar Logistics — Daily Revenue Validation

Automates routine financial data-quality assurance over source and summarised revenue datasets without relying on an LLM for exact calculations.

- Implemented scheduled or manually initiated n8n checks backed by PostgreSQL SQL queries for completeness, nulls, duplicate records and customer coverage.
- Compared source and aggregated revenue values for reconciliation, marking passing datasets as ready for downstream processing.
- Issued Slack exceptions on failed checks, deliberately using **deterministic SQL rather than AI** for repeatability and auditability.

Northstar — Travel Exception Approval

Demonstrates a controlled approval process where AI interprets an employee's justification but cannot independently authorise an exception.

- Accepted requests through an n8n form, validated structured inputs and calculated policy-relevant facts.
- Used OpenAI to summarise or interpret justifications, while deterministic policy rules established approval authority and routing.
- Integrated Slack review with execution pause/resume and human APPROVE/REJECT capture; recorded decisions and notified requesters.
- Controlled prototype using simulated policy inputs; manager escalation, persistent enterprise audit storage and enterprise authentication are not claimed as implemented.

ADDITIONAL SELECTED AI ENGINEERING PROJECTS

Career Intelligence Copilot (CIC)

Python · Pydantic · OpenAI APIs · PyTest · Git

Built a decision-support system for a real AI Engineering job search, helping assess opportunities, decide which applications to pursue and prepare evidence-grounded application materials while leaving decisions with the user.

- Modelled a structured career profile and opportunity evidence, producing explainable assessment and application strategies with provenance.
- Combined LLM interpretation and drafting with deterministic workflow states, duplicate detection, validation and explicit owner review.
- Used regression/golden scenarios, acceptance gates and iterative calibration to improve reliability; outputs still require human review.

Governance-Aware Document Intelligence RAG

Python · FastAPI · Docker · LangChain · ChromaDB · OpenAI · PyTest

Designed document-question answering for organisational policies, procedures and other internal knowledge, emphasising grounded answers and visibility into unsupported responses.

- Implemented ingestion of text, PDF and DOCX documents, embedding generation, retrieval and cited response construction.
- Introduced telemetry, evaluation and grounding controls so uncertain or unsupported responses could be identified rather than presented as established facts.
- Separated API/service functions from the user-facing experience, with automated tests of core behaviour.

Operational Intelligence Copilot

Python · FastAPI · OpenAI APIs · PyTest

Built an analytics-led assistant for exploring business operations and explaining trends, variances, anomalies and risks using evidence from underlying datasets.

- Combined dataset profiling, deterministic calculations and analysis workflows with controlled LLM narrative generation.
- Kept quantitative findings in conventional analytic logic, using AI to make validated results easier for decision-makers to interpret.
- Regression-tested analytical outputs and structured explanations for traceability.

Payroll Diagnostics Engine & AWS Deployment

Python · Pandas · Docker · S3 · ECR · ECS/Fargate · IAM · CloudWatch · PyTest

Engineered a rule-based payroll assurance workflow that examines workforce/payroll records for anomalies, reconciles inputs and creates explainable findings and executive reporting.

- Created deterministic diagnostics and reproducible report-generation pipelines, with controls to distinguish verified findings from possible anomalies.
- Migrated containerised processing from local execution to AWS using ECR, ECS/Fargate and client-scoped S3 storage with IAM and CloudWatch visibility.
- Compared cloud execution with a frozen local baseline, exercising synthetic success/failure scenarios and client-isolation checks.

Public Holiday Entitlements Application

Python · Streamlit · SQLite · REST APIs · geospatial processing

Built an Australian public-holiday entitlement tool that uses location and applicable rules to determine relevant holiday information for payroll and HR workflows.

- Combined geocoding, local-government-area mapping and public-holiday API data with deterministic entitlement logic.
- Supported individual and batch inputs, with downloadable compliance-oriented reports.

COMMERCIAL ENGINEERING EXPERIENCE

Data Engineer — nbn Australia

Mar 2020 – Oct 2023

- Developed and supported enterprise data pipelines and operational reporting with AWS, Python, SQL and Apache NiFi; used Apache Airflow for orchestration.
- Applied PostgreSQL, data validation, automated testing, Git/Jenkins CI/CD and Agile delivery in production engineering environments.

Earlier Commercial Roles — Test Analyst / Test Automation Engineer

Worked across enterprise software delivery, automation frameworks, regression testing and production support at Bakers Delight, Console and AccessHQ.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

AWS Certified Developer – Associate (expired) · Databricks Data Engineer Professional (expired)

Professional development (Oct 2023 – Nov 2025): structured Data Engineering and Applied AI study, including Python, LLM application development, LangChain and cloud/data platforms. General Assembly Data Science Immersive (2019).